Yuni Wu

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Summary

Robotics AI Engineer focused on making autonomy decisions measurable under physical-world ambiguity. Turns Go1 telemetry, VLM decisions, and behavior-cloning outputs into reproducible evaluation loops for robot behavior.

Experience

Graduate Research Assistant Sept 2025 – May 2026
CAIRO Lab, University of Colorado Boulder — Embodied AI and robotics research lab

- Curated 97K+ Unitree Go1 state–action samples with ROS/rosbag pipelines for imitation-learning evaluation.
- Constructed and benchmarked an offline evaluation dataset containing 2.4K+ labeled state–action sequences across 13 robot-navigation scenarios to assess VLM decision-making consistency.
- Benchmarked VLM-guided navigation decisions for consistency and deployment robustness.

Machine Learning Engineer Intern May 2025 – Aug 2025
MyEdMaster LLC — AI healthcare and education startup

- Engineered a PyTorch/torchvision ResNet workflow on 50K+ image samples, achieving 0.92 F1-score.
- Reduced inference latency by 5$\times$ (0.15s to 0.03s per image) through ResNet ablation and deployment tradeoff analysis.
- Built reliability checks with cross-validation, ROC-AUC, and calibration to measure model behavior under class imbalance.

Software Engineer June 2021 – Aug 2022
Shopee Co. Ltd. — Leading Southeast Asian e-commerce platform

- Reduced QA workload by 80% by engineering SQL validation and anomaly-detection pipelines on 800+ QA scenarios.
- Accelerated inconsistency detection with regression monitoring workflows for production reliability analysis.

Projects

VLM-Based Social Navigation for Quadruped [project]

- Evaluated VLM-based social-navigation behavior on a Unitree Go1 quadruped for human-aware navigation evaluation.
- Benchmarked InternVL, Qwen-VL, and prompt strategies across 2.4K+ labeled state–action sequences and 13 scenarios.
- Achieved 0.11 MAE and $R^2 = 0.62$ by training PyTorch/torchvision ResNet policies on 97K+ samples.

NailAI — Cloud-Native Vision Inference [demo]

- Evaluated cloud-native image classification with PyTorch/torchvision ResNet inference workflows.
- Assessed event-driven Cloud Run, Pub/Sub, and Cloud Storage architectures for asynchronous inference reliability.
- Achieved 0% failure at 200 concurrent users with p95 latency of 6.8s through fault-tolerant orchestration.

Technical Skills

Languages: Python, SQL, Go, R
Robotics AI: ROS, rosbag, Unitree Go1, VLM evaluation, behavior cloning, waypoint prediction, scenario diagnostics
Machine Learning: PyTorch, torchvision, ResNet, CNNs, Transformers, offline evaluation, benchmarking, residual diagnostics
Evaluation: MAE, R-squared, F1-score, cross-validation, calibration, ablation workflows, deployment robustness
Systems: Cloud Run, Pub/Sub, Cloud Storage, Docker, data pipelines, monitoring, validation workflows

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0 May 2026
Soochow University — BS, Data Science | GPA: 3.7/4.0 June 2021